Optimization: Critical Points & Lagrange Multipliers

Finding critical points

1 Find the critical point(s) of $f(x, y) = x^2 + 3y^2 - 4x + 6y$.

2 Find the critical point(s) of $f(x, y) = xy - x^2 - y^2$.

The Hessian and the Second Derivative Test

3 Classify the critical point $(2, -1)$ of $f(x, y) = x^2 + 3y^2 - 4x + 6y$ using $D = f_{xx}f_{yy} - (f_{xy})^2$.

4 Classify the critical point $(0, 0)$ of $f(x, y) = xy - x^2 - y^2$.

5 Classify the critical point of $f(x, y) = x^2 - y^2$.

Lagrange multipliers: setting up the system

6 Set up the Lagrange multiplier equations to optimize $f(x, y) = xy$ subject to $g(x, y) = x + y - 10 = 0$.

Solving Lagrange multiplier problems

7 Maximize $f(x, y) = xy$ subject to $x + y = 10$.

8 Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.

A geometric Lagrange multiplier application

9 Find the point on the line $x + y = 6$ closest to the origin, using Lagrange multipliers on $f(x, y) = x^2 + y^2$.